

Modeling Kinesin Motility Using a Sloping Potential and Stochastic Chemical Transitions

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Kinesin is a processive molecular motor that transports cargo along microtubules through discrete, ATP-driven steps. In this work, we present a minimal chemo-mechanical model in which each kinesin head undergoes stochastic chemical state transitions while moving along a sloping periodic potential representing the microtubule lattice. The model reproduces essential features of kinesin hand-over-hand stepping, including ~ 8 nm step sizes, occasional backward and inchworm steps, and power-stroke-like forward motion of the detached head driven by differential binding affinities. Inter-head interactions prevent overlap and excessive stretching, supporting processivity without enforcing explicit coordination. Simulation results at 1 mM ATP show mean velocities consistent with experimental observations (~ 400 nm/s), demonstrating that directional motion can emerge from the interaction between a biased energy landscape and stochastic chemical transitions. This framework demonstrates how minimal mechanical and chemical components can capture kinesin motility.

Project Topic: How do molecular motors do work?

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I. INTRODUCTION

Kinesin is a processive molecular motor that moves unidirectionally, in a hand-over-hand fashion [1], along microtubules in discrete steps of approximately 8 nm, powered by ATP hydrolysis induced conformational changes [2]. Processivity refers to the ability of a motor protein to remain attached to its filament while executing multiple successive steps before detaching [3]. Microtubules are hollow cylindrical polymers composed of α and β -tubulin dimers that form a major component of the eukaryotic cytoskeleton and serve as tracks for intracellular transport. Kinesin's stepping behavior arises from ATP-driven conformational changes in its motor heads, which produce a power stroke: a mechanical movement that propels the attached head forward along the microtubule. In addition, the periodic binding potential of the microtubule lattice guide the motor's motion, while thermal fluctuations enable stochastic transitions between microtubule binding sites [4] [5] [6].

In this project, Kinesin motility is modeled using an effective potential framework. The microtubule track is represented as a downward-sloping potential landscape with regularly spaced wells separated by approximately 8 nm, corresponding to Kinesin binding sites [5]. The interaction between the two motor heads is incorporated through an inter-head potential that is repulsive at short separations and attractive at larger distances, enforcing coordinated stepping. Thermal fluctuations are included using Langevin dynamics, allowing stochastic forces to drive transitions between wells, while chemical state transitions associated with ATP hydrolysis modulate the effective potentials experienced by the motor.

Mechanisms such as the Brownian ratchet illustrate how random thermal motion can be rectified into direc-

tional motion through energy landscape modulation [6]. Although the model used here is not a strict Brownian ratchet, the sloping potential landscape serves a similar purpose: it biases stochastic motion in the forward direction, producing discrete steps along the microtubule [7]. This combined chemo-mechanical description captures essential features of Kinesin motion and demonstrates how directional stepping can emerge from a stochastic system with energy input.

II. OVERVIEW

Several modeling approaches have been developed to describe kinesin stepping, differing in how they balance chemical kinetics, mechanical motion, and stochastic effects. Existing models include Brownian ratchet frameworks that emphasize the rectification of thermal fluctuations through ATP-dependent modulation of the energy landscape, Markov chain models that describe stepping as transitions between discrete chemical and mechanical states, and Brownian dynamics models that explicitly resolve spatial motion along the microtubule using effective potentials. Each approach captures different aspects of kinesin motility and involves distinct assumptions regarding force generation, conformational change, and stochasticity.

Method 1: Brownian Ratchet Model. This method models kinesin motion as arising from thermal fluctuations that are rectified by an asymmetric energy landscape. The detached motor head undergoes Brownian motion along the microtubule, while periodic changes in the potential, driven by ATP binding and hydrolysis, alternately allow and restrict forward and backward diffusion. Directional stepping emerges from the time-dependent modulation of binding affinities, rather than

a power stroke, which favors forward attachment over backward binding. This approach captures the stochastic nature of motor motion with minimal mechanical assumptions.

Method 2: Pure Markov Chain Model. In this method, kinesin stepping is modeled as a sequence of discrete chemical and mechanical states connected by transition rates. Each state represents a specific configuration of head attachment and nucleotide binding, and stepping occurs through probabilistic transitions governed by experimentally defined rates. Spatial motion is not modeled explicitly; instead, forward movement is inferred from transitions between states corresponding to successive microtubule binding sites. This approach provides a compact and computationally efficient description of kinesin dynamics, focusing entirely on state kinetics rather than force generation.

Method 3: Brownian Dynamics with Potential Wells. This method treats the detached Kinesin head as a particle undergoing stochastic thermal motion (Brownian motion) along the microtubule. The microtubule binding sites are modeled as potential wells, representing energetically favorable positions for the head to attach. ATP binding to the leading head triggers neck linker structural change, which shifts the energy landscape and biases the free head toward the forward site. This approach summarizes both the mechanical motion and the ATP-driven directional bias, making it suitable for a realistic simulation of Kinesin stepping.

Among the three approaches, Method 3 (the Brownian dynamics model with potential wells) was selected because it balances mechanical realism and chemical bias. Unlike the Brownian ratchet model (Method 1), it allows explicit representation of the kinesin head's spatial motion while still capturing stochastic fluctuations. Compared to the pure Markov chain model (Method 2), it incorporates force generation and conformational changes, providing a more physically grounded simulation of stepping. The sloping potential effectively biases thermal motion in the forward direction, reproducing directional stepping while remaining computationally tractable, making it the most suitable framework for our project. Refer to table I for a summary of the above mentioned methods.

III. METHOD

The Kinesin motor is modeled as a two-headed particle system moving along a one-dimensional coordinate representing the microtubule axis. Each head undergoes stochastic motion under the influence of effective potentials, inter-head interactions, and thermal noise. Directed motion emerges from the coupling between a spatially sloping potential landscape and stochastic chemical state transitions associated with the ATP hydrolysis cycle.

A. Microtubule Potential Landscape

The microtubule is represented by a spatially periodic potential with minima separated by approximately 8 nm, corresponding to Kinesin binding sites along the microtubule. To bias motion in the forward direction, the periodic potential is superimposed on a constant downhill slope, representing the free-energy bias introduced by ATP hydrolysis. The energy landscape could be seen in Figure 1.

The resulting effective potential experienced by each head is given by the binding affinity

$$V_b(x) = V_{b0} \sum_i \left[-e^{-(x-r_i)^2} \right] - sk_b T x \quad (1)$$

where V_{b0} is a constant that depends on the chemical state of the Kinesin head, whose values are $7k_bT$, $10k_bT$ and $16k_bT$, respectively, for ADP bound, empty and ATP bound heads. x is the position of the Kinesin head along the microtubule, r_i is the location of the i -th binding site along the microtubule, s is the slope of the potential landscape whose value is 0.31 that imparts the downward slope along increasing x , k_b is the Boltzmann constant and T is the ambient temperature taken to be 300K. In equation (1), the summation is over all binding sites.

Classical Brownian ratchet models describe kinesin motion through time-dependent switching between distinct asymmetric energy landscapes, in which thermal fluctuations are rectified by ATP-driven changes in binding affinity. In contrast, in the present model, directionality is introduced through a constant global tilt superimposed on a periodic binding potential, providing a steady energy bias toward forward motion. While this sloping potential does not constitute a Brownian ratchet in the strict sense, it serves a similar role by biasing stochastic thermal motion and enabling directed stepping. This formulation offers a simplified yet physically motivated representation of how chemical energy from ATP hydrolysis is converted into mechanical motion.

The sloping potential well was developed independently in this work and was later found to be consistent with the modeling approach presented in [7]. However, the implementation adopted here differs substantially and is considerably simpler.

B. Interhead Interaction Potential

Mechanical coordination between the two Kinesin heads is enforced through an inter-head interaction potential that depends on their separation $|x_1 - x_2|$. The interaction is repulsive at short distances to prevent overlap and attractive at larger distances to maintain tethering between the heads.

The first force is effective when the two heads are separated by $> 9\text{nm}$ with a potential

Model	Use case scenario	Features	Suitable for the project?
Brownian Dynamics with Potential Wells	Modeling Kinesin stepping along microtubules	- Free head undergoes stochastic Brownian motion - Microtubule represented as periodic potential wells	Yes: captures both mechanical motion and ATP-driven directional bias
Brownian Ratchet Model	Explaining directional motion arising from thermal fluctuations	- Motion generated by Brownian diffusion in an asymmetric, time-dependent potential - Directionality produced by modulation of binding affinities - No explicit power-stroke mechanism	Partially — captures stochastic directionality but lacks explicit mechanical force generation
Pure Markov Chain Model	Kinetic description of kinesin stepping cycles	- Discrete chemical and mechanical states - Stepping described by probabilistic state transitions - No explicit spatial or mechanical modeling	No — efficient for kinetics but does not represent physical motion or forces

TABLE I. Comparison of modeling approaches for simulating Kinesin motility, use cases, key features, and suitability for the project.

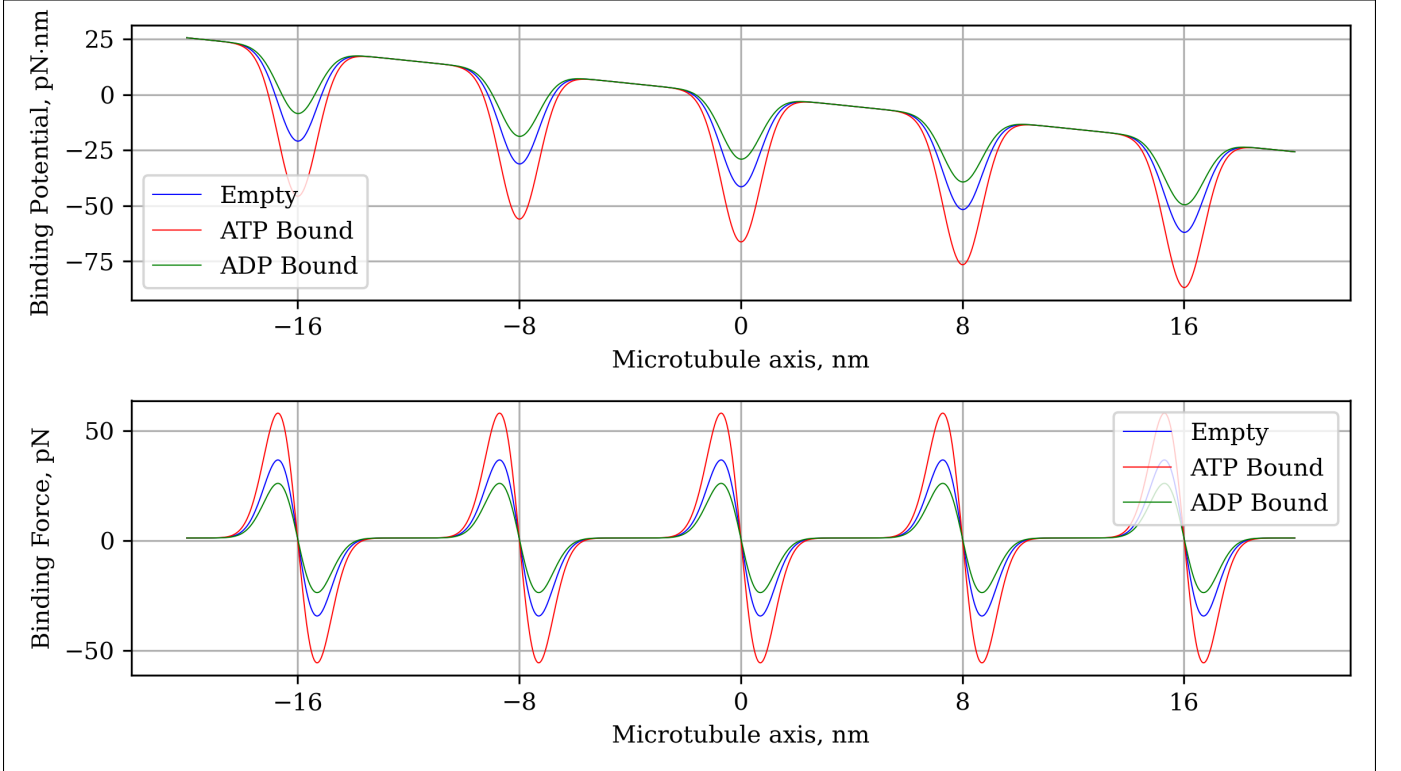


FIG. 1. **Sloped periodic potential model.** Schematic of the effective potential landscape used to model kinesin motion along a microtubule. Periodic potential wells represent discrete microtubule binding sites, spaced approximately 8 nm apart, corresponding to the kinesin step size. A constant downhill slope is superimposed on the periodic potential, introducing a directional energy bias that favors forward motion. Thermal fluctuations enable stochastic transitions between adjacent wells, while the global tilt breaks symmetry and produces net directed transport. The three curves correspond to nucleotide-free, ADP-bound, and ATP-bound kinesin heads, with the ADP-bound state exhibiting the weakest binding affinity and the ATP-bound state the strongest. As a result, the ADP-bound head is more likely to detach during the ATP hydrolysis cycle, enabling alternating head attachment and producing the characteristic hand-over-hand walking of kinesin along the microtubule.

$$V_a(x) = V_{a0}e^{|x_1-x_2|-9} \quad (2)$$

and the second force is effective when the two heads get closer, with a potential

$$V_r(x) = V_{r0}e^{-(x_1-x_2)^2} \quad (3)$$

where V_{a0} is taken as $10k_bT$ and V_{r0} is taken as $70k_bT$. Figure 2 illustrates the effect of each of these potentials on the individual Kinesin heads. This inter-head interaction, together with the periodic potential well structure (equation 1), is expected to maintain the characteristic 8 nm step length of kinesin motion.

This choice of interhead interaction potential is motivated by the effective potential described in [5], although

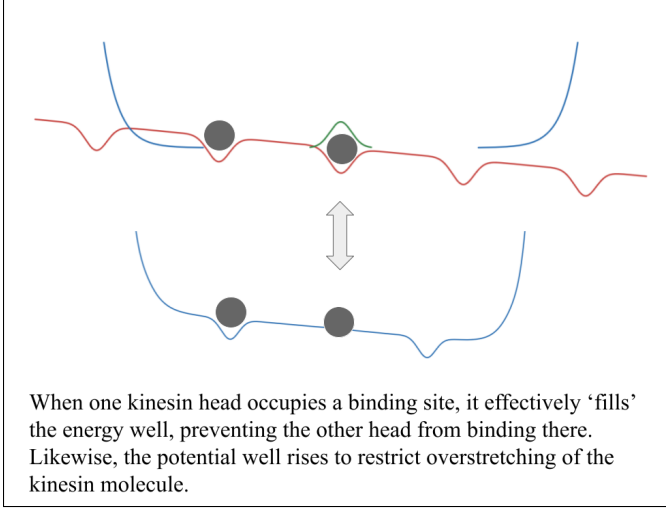


FIG. 2. **Schematic representation of the effects inter-head interaction.** The attractive inter-head potential is modeled as a symmetric function that rises exponentially with distance on either side of a head, penalizing large separations between the two heads. The short-range repulsive component creates an inverted potential well of finite depth. As a result, the effective binding potential experienced by an incoming head is significantly reduced at sites already occupied by the other head, making binding to an occupied location highly unlikely. These potentials act symmetrically, with each head experiencing the same interaction due to the presence of the other.

the present model differs in implementation.

C. Langevin Dynamics

The motion of each Kinesin head is governed by the finite difference equation from Langevin dynamics [8]

$$x_{1,i} = x_{1,i-1} + \frac{F_{1,i}}{\gamma} \Delta t + \Delta x_1 \quad (4)$$

and

$$x_{2,i} = x_{2,i-1} + \frac{F_{2,i}}{\gamma} \Delta t + \Delta x_2 \quad (5)$$

Δx_1 and Δx_2 are random displacements due to stochastic forces. As the manifestation of thermal white noise, they can be written as

$$\Delta x_1 = \sqrt{2D\Delta t} w_{1,i} \text{ and } \Delta x_2 = \sqrt{2D\Delta t} w_{2,i} \quad (6)$$

where $w_{1,i}$ and $w_{2,i}$ are independent Gaussian random numbers with a zero mean and unit standard deviation. D is the diffusion co-efficient with value taken to be $0.1 \text{ nm}^2/\mu\text{s}$ and γ is the friction co-efficient, given by $k_b T/D$ [8].

$F_{1,i}$ and $F_{2,i}$ are the resultant forces on the individual heads arising out of the microtubule potential and interhead interaction potentials, given by

$$F = - \left(\frac{dV_b}{dx} + \frac{dV_a}{dx} + \frac{dV_r}{dx} \right) \quad (7)$$

D. Chemical State Dynamics

Each kinesin head is assigned a discrete chemical state representing its nucleotide-binding condition (ATP-bound, ADP-bound, or nucleotide-free). Transitions between these states are modeled as stochastic jump processes with state-dependent rates, treated as Poisson processes and simulated using a kinetic Monte Carlo scheme. The time until the next chemical transition is determined using the standard Poisson waiting-time formula:

$$\Delta t = - \frac{\ln u}{k} \quad (8)$$

where $u \in (0, 1)$ is a uniformly distributed random number and k is the rate of the specific transition. In this simulation, three rates are considered: the ATP-binding rate k_{ATP} , the ADP-release rate k_{ADP} , and the ATP-hydrolysis rate k_{hyd} . This approach ensures that chemical transitions occur randomly, but with the correct average frequency dictated by each rate [9].

Importantly, no chemical gating or position-dependent coordination between the two heads is imposed, unlike in other studies [5], as the present model intentionally focuses on a minimal chemo-mechanical description. Chemical transitions occur independently of the mechanical state of the motor, and are not conditioned on head attachment, detachment, or inter-head proximity. As a result, ATP binding, hydrolysis, and ADP release may occur in either head regardless of the state of the partner head.

The values of the ATP-binding rate k_{ATP} , the ADP-release rate k_{ADP} , and the ATP-hydrolysis rate k_{hyd} are taken to be $0.4 \text{ M}^{-1}\mu\text{s}^{-1}$, $200 \times 10^{-6} \mu\text{s}^{-1}$ and $600 \times 10^{-6} \mu\text{s}^{-1}$ respectively.

E. Chemical-Mechanical Coupling

ATP binding increases the binding affinity of the bound head to the microtubule, while the ADP-bound head is weakly bound and can be displaced by thermal fluctuations. Once the ADP-bound head detaches, it undergoes stochastic diffusion along the microtubule, biased forward by the sloping potential landscape (equation 1). The inter-head interaction (equations 2 and 3) prevents the detached head from rebinding to the leading head's site, directing it toward the next available binding site. This coordinated interaction between a tightly

bound ATP head and a diffusing ADP head produces motion analogous to the kinesin power stroke. Together, the stochastic chemical transitions and mechanically biased diffusion capture the characteristic stepping behavior of kinesin (see Figure 3 for an illustration of the ATP cycle driving kinesin stepping).

F. Programmatic Implementation

The model parameters V_0 , s , V_{a0} , V_{r0} , D , k_{ATP} , k_{ADP} , and k_{hyd} are chosen to reproduce an average kinesin stepping speed of approximately 400 nm/s [5] under no external load, consistent with experimental observations.

In Equation 1, the summation is formally over all binding sites (modeled as infinite in number). In the program implementation, however, the sum is restricted to the four nearest sites, as contributions from more distant sites are negligible and limiting the sum reduces computational cost.

To prevent division by zero, a very small value (10^{-50}) is added to the divisor wherever a division occurs. The program is implemented in Python, with NumPy used for all numerical computations. Vectorized operations and bulk random number generation are employed wherever possible to improve performance, and Matplotlib is used for plotting figures.

IV. RESULTS AND DISCUSSION

The results presented in this section are obtained from an ensemble of independent simulations. For each parameter set, 100 simulation runs were performed to account for stochastic variability arising from thermal noise and chemical state transitions. The instantaneous position of the kinesin motor was defined as the average position of the two heads along the microtubule axis. The mean velocity for each run was computed as the net displacement of this averaged position over the total simulation time, and ensemble statistics were obtained by averaging over all runs.

A. Emergence of Directed Motion

The coupled chemo-mechanical model produces sustained, directional motion of the kinesin motor along the microtubule axis. Although thermal fluctuations are strong, the motor exhibits a net forward displacement over time, showing that directional motion arises naturally from the interactions between the forward-sloping potential landscape and stochastic chemical state transitions. The sloping potential biases detached heads to move preferentially in the forward direction once they escape their binding wells, while inter-head interactions prevent rebinding to occupied sites, ensuring hand-over-

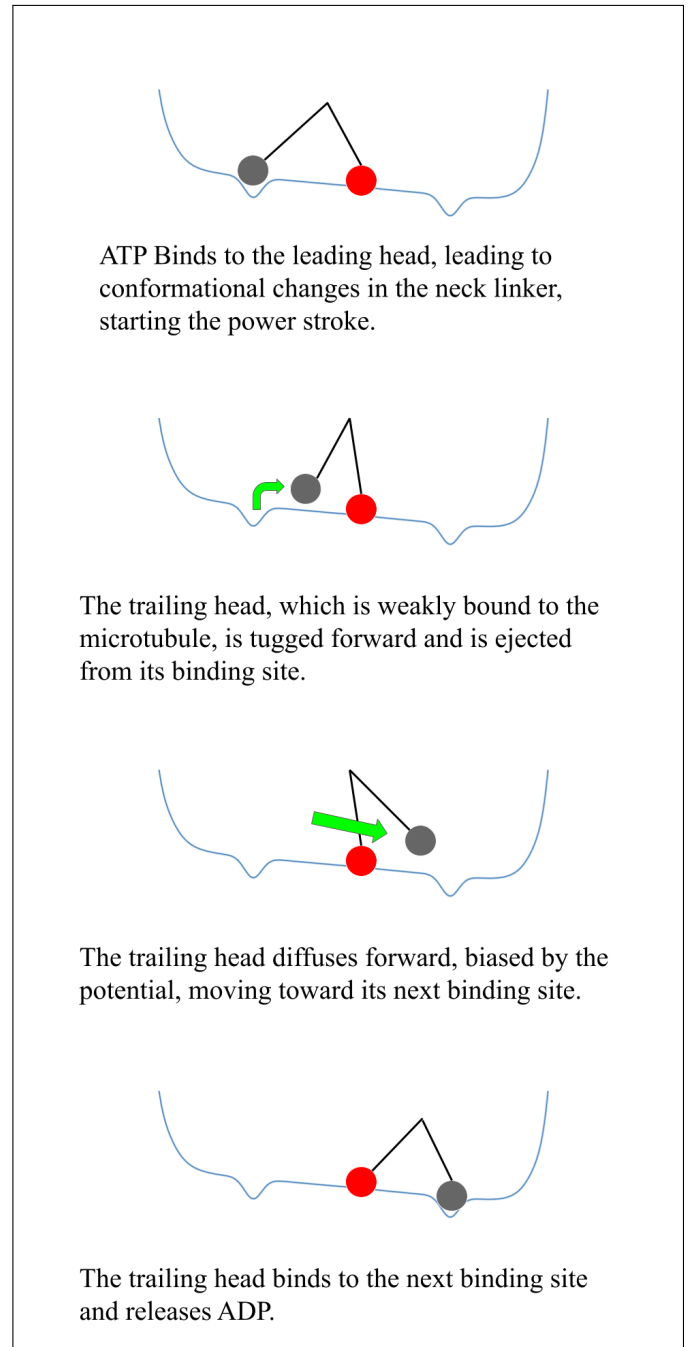


FIG. 3. The ATP cycle and its relationship to Kinesin stepping mechanism. ATP binding and hydrolysis in the kinesin heads modulate the depths of the binding potential wells, causing the ADP-bound head, which is weakly bound, to detach from its site due to thermal fluctuations. The forward-sloping potential landscape biases the motion of the detached head toward the next binding site, simulating the power stroke. Meanwhile, the ATP-bound head remains tightly attached, with thermal noise insufficient to displace it.

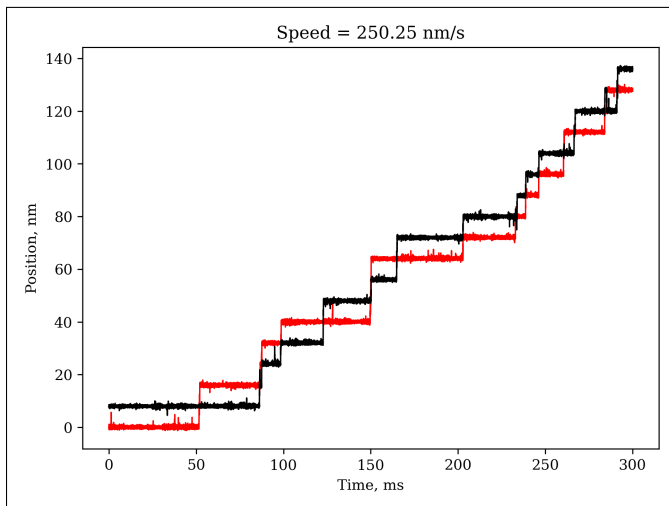


FIG. 4. **Kinesin trajectory.** The figure shows one trajectory from the simulation, under an ATP concentration of 1 mM. The simulation was run 100 times and the corresponding speed distribution is shown in Figure 5. The motion exhibits a substantial amount of hand-over-hand stepping, as well as forward movement that occurs through inchworm transitions. These inchworm steps arise primarily from the stochastic nature of the ATP cycle in both heads, where ATP cycling coordination is not strictly enforced, and therefore these events do not faithfully replicate the mechanism illustrated in figure 3 for every step.

hand progression. These results are illustrated in Figures 4, 5 and 7.

ATP-driven changes in binding affinity further modulate this motion: ADP-bound heads are weakly attached and can detach due to thermal noise, whereas ATP-bound heads remain tightly bound, effectively producing a power-stroke-like mechanism that coordinates motion between the two heads. At an ATP concentration of 1 mM, simulations over multiple runs yield a mean kinesin velocity of approximately 400 nm/s, in close agreement with experimental measurements.

In simulations where either the downhill slope is removed (Figure 8) or the ATP cycle is disabled (Figure 7), the motor reduces to unbiased diffusion along the microtubule with statistically negligible net drift, confirming that directed motion in this model depends on the combined chemo-mechanical bias.

B. Stepping Behavior and Step Size

The kinesin motor advances predominantly in discrete steps of approximately 8 nm, corresponding to transitions between neighboring potential wells in the microtubule landscape. This step size arises naturally from the spacing of the binding sites encoded in the sloping periodic potential. The distribution of step sizes is sharply peaked around 8 nm, reflecting the periodic lattice structure. Refer Figure 4 for the forward processive motion

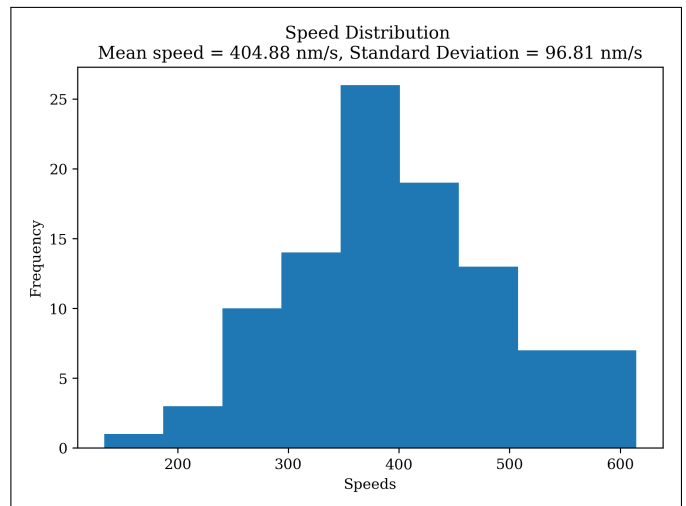


FIG. 5. **Speed distribution of Kinesin motion.** The figure shows the distribution of kinesin walking speeds along a microtubule in an aqueous environment at an ATP concentration of 1 mM. The distribution is approximately normal, with a mean speed that closely matches the experimentally observed value of about 400 nm/s.

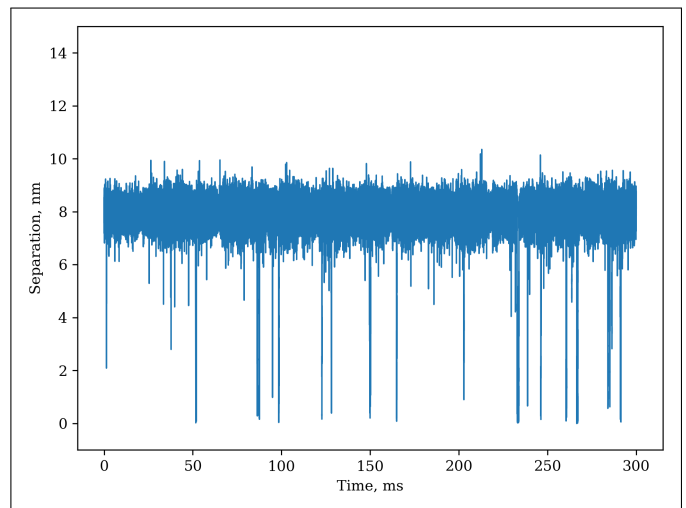


FIG. 6. **Separation of Kinesin heads over time.** The picture shows separation between Kinesin heads corresponding to the trajectory in Figure 4. During the simulation, the two kinesin heads maintain an approximate 7–9 nm spacing, enforced by the inter-head interaction potentials and the periodic binding wells of the microtubule lattice. Thermal fluctuations generate Brownian motion, and occasional drops to zero indicate the stepping events of the heads.

and Figure 6 for the separation between heads during the time of motion.

Occasional backward steps are observed due to thermal activation (not shown in any of the plots), highlighting the stochastic nature of stepping at the molecular scale. Forward steps occur more frequently than backward steps because of the combined effect of the forward-sloping potential and chemical-state-dependent modulation of the

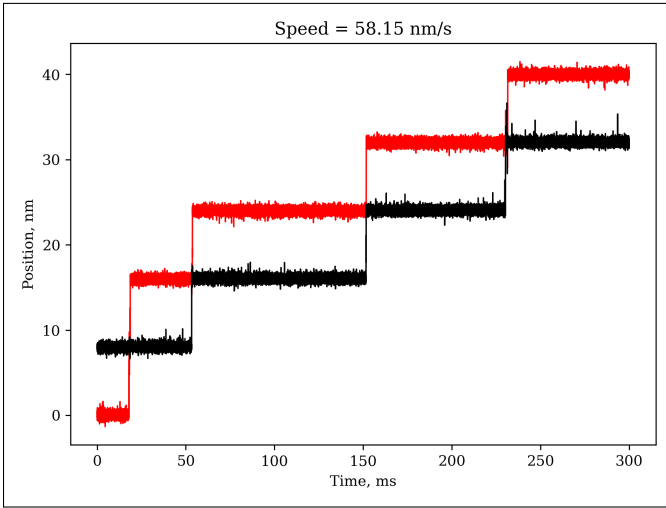


FIG. 7. **Kinesin trajectory in the absence of ATP.** The figure shows the kinesin trajectory at zero ATP concentration, corresponding to the absence of an ATP hydrolysis cycle. Without an energy source to drive stepping, the kinesin remains largely stationary on the microtubule, exhibiting only minimal motion due to thermal fluctuations.

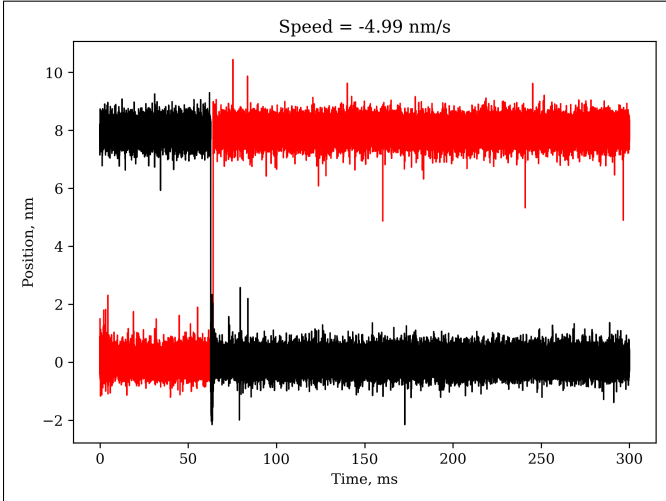


FIG. 8. **Kinesin trajectory on a flat (zero-slope) landscape.** The figure shows the positions of the two kinesin heads over time when the forward-sloping potential is removed. Without a directional bias, the heads remain near their initial positions for extended periods, with occasional exchanges of position driven solely by thermal fluctuations. This illustrates that in the absence of a slope, no net forward motion emerges and stepping is purely stochastic. Negative velocity in this plot arise from head-swapping event in which the two kinesin heads exchange positions without net forward translation of the motor. Since the motor position is defined as the average of the two head positions, such exchanges can produce an apparent backward displacement, while the actual displacement, and consequently the velocity, is zero.

binding wells: ADP-bound heads detach easily and diffuse preferentially forward, while ATP-bound heads remain tightly bound, producing a power-stroke-like effect that biases motion in the intended direction.

Because chemical transitions in the two heads are independent and uncoordinated, the motor sometimes exhibits inchworm motion in addition to the standard hand-over-hand stepping (see Figure 3 for an illustration of hand-over-hand stepping mediated by the ATP cycle). These irregular steps arise when the ATP cycle in one head is out of sync with the other, causing the free head to advance in a way that does not strictly follow alternating stepping.

Overall, the interactions of thermal fluctuations and ATP-driven chemical transitions ensures that the motor exhibits net processive motion along the microtubule while maintaining the characteristic ~ 8 nm step size observed experimentally [10].

C. Role of Inter-Head Interaction

The inter-head interaction potential regulates the relative positions of the two kinesin heads. Short-range repulsion prevents both heads from occupying the same binding site, while longer-range attraction limits excessive stretching and keeps the heads tethered near each other (refer Figure 6). Weakening or removing this interaction leads to occasional overlap, excessive separation, and reduced processivity.

D. Influence of Thermal Noise

Thermal fluctuations enables barrier crossing between adjacent potential wells, facilitating stepping events. On longer time scales, excessive thermal noise increases the probability of backward steps and reduces directional efficiency. An intermediate noise level maximizes forward velocity, consistent with Brownian ratchet-like behavior where stochastic forces are essential for motion but must be rectified by the energy landscape. In this model, a diffusion coefficient of $D = 0.1 \text{ nm}^2/\mu\text{s}$ is found to provide the optimal balance for consistent and unidirectional Kinesin motion.

E. Effect of Stochastic Chemical State Transitions

Repeated simulations indicate that the mean kinesin velocity increases with ATP concentration. However, this increase is accompanied by large standard deviations, particularly at higher ATP levels (refer Figure 9). This substantial variability suggests that, while higher ATP availability can promote forward motion on average, the stepping dynamics remain highly stochastic in the absence of inter-head chemical gating.

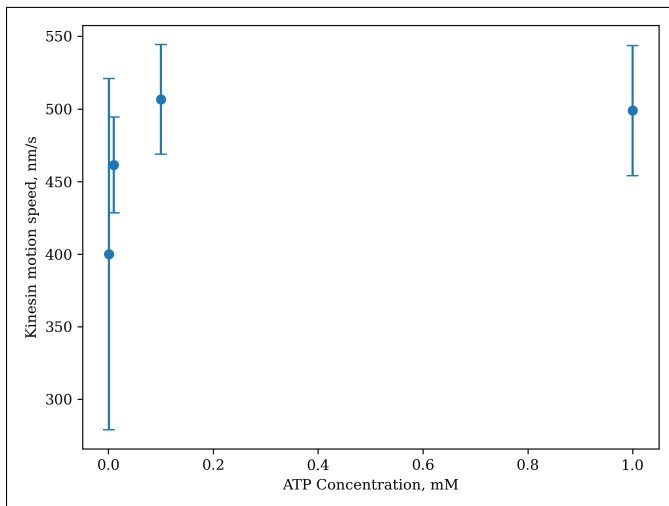


FIG. 9. ATP concentration - Kinesin velocity relationship. The figure shows the mean kinesin velocity as a function of ATP concentration, with error bars indicating the standard deviation over multiple simulation runs. While the mean velocity increases with ATP concentration, the large variability at higher ATP levels indicates highly stochastic stepping dynamics. This behavior arises from the absence of inter-head chemical gating in the model, leading to intermittent trapping in strongly bound ATP states and irregular stepping, rather than stable, saturating motion.

The large spread in velocities reflects frequent trapping in strongly bound ATP states as well as occasional rapid stepping events when a weakly bound head is present. As a result, the apparent increase in mean velocity does not correspond to a consistent processive motion due to increased ATP concentration.

Overall, the results suggest that without coordination between the ATP cycles of the two heads, increased ATP concentration leads to unstable and highly variable motility rather than robust, saturating transport, underscoring the role of chemical gating in reducing fluctuations and stabilizing kinesin’s walking speed.

V. CONCLUSIONS AND OUTLOOK

This work demonstrates that processive, directional motion of kinesin can emerge from a minimal chemo-mechanical model in which stochastic chemical transitions and a sloping periodic potential together bias head movement. The simulations show that discrete stepping along the microtubule, including both forward and occasional backward or inchworm steps, can be reproduced without enforcing explicit coordination between the heads.

The model highlights how the combination of chemical-state-dependent modulation of binding potentials and a forward energy bias produces power-stroke-like behavior and net forward motion. This indicates that stochastic chemical transitions, when coupled with an appropriately

biased energy landscape, are sufficient to generate processive transport along the microtubule.

Finally, the results underline the importance of chemical modulation and the sloping potential in stabilizing motion. Even without coordinated ATP cycling or head gating, these minimal ingredients reproduce several qualitative features of kinesin walking observed experimentally, providing a framework for understanding directional motility.

A natural next step for this model is to incorporate inter-head chemical gating, which would coordinate the ATP cycles of the two heads and promote consistent hand-over-hand stepping, reducing inchworm steps observed in the current ungated model. Another avenue for refinement is the mechanical representation of kinesin: most models treat the motor as a spring that stores energy to move the trailing head during ATP hydrolysis, whereas in this work, forward motion arises from the sloping potential combined with weaker binding in the ADP state. Exploring a hybrid approach or more detailed mechanical models could provide further insight into the interactions between chemical and mechanical forces.

In addition, the repulsive inter-head potential could be modified to act only near binding sites, rather than continuously, allowing closer head approaches without destabilizing motion. Once these refinements: gating, mechanical model adjustments, and repulsive potential modification, are implemented, parameter sensitivity studies could be performed, systematically varying ATP binding and hydrolysis rates, potential depths, and diffusion coefficients to quantify their effect on stepping behavior and velocity.

Together, these extensions would enhance the realism of the model and provide a flexible framework to study kinesin motility under diverse conditions.

VI. CONTRIBUTIONS

Abhishek Udupa and Annamalai Vellaichamy conceived the study, designed and performed the experiments, analyzed the data and wrote the manuscript. All authors reviewed and approved the final manuscript.

VII. CONFLICT OF INTEREST

The authors declare no known conflicts of interest related to this work. Any inadvertent or unrecognized conflicts are unintended and are disclosed here for transparency. This work did not receive any specific grant or external funding. All research, simulations, and analyses were conducted independently.

VIII. DATA AND CODE AVAILABILITY

The code used to run the simulations and generate the figures in this study is publicly available at <https://github.com/abhishekaudupa/kinesin-processivity>.

The repository includes all scripts, parameters, and example data used in this study and is released under an MIT license. All data presented in this report are generated by running the program: no pre-generated data files are included in the repository.

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